

# TOPIC

Detection Of Jihadism In Social Network  
Using Big Data Techniques Supported By  
Graphs And Fuzzy Clustering

# contents



1. Introduction
2. Description of methodology
3. Result and discussion
4. Data availability
5. Application



## 2. Description of methodology architecture

1. Big data architecture
2. Fuzzy architecture to isolate the suspicious profiles
3. Tweet extraction technique
4. Information preprocessing
5. Graphing the network
6. Fuzzy segmentation

# Technique Used Here..

- Social network analysis.
- Sentiment analyze of messages.
- Graph (uses impact on each user in network)
- Fuzzy clustering techniques( classify users in profiles)

# steps...

1. Acquire information from Kaggle's database and twitter user's accounts
2. filtered less and most active users (generating info And retweeted info from other)
3. Sentiment analysis of the messages(using different parameter from the msg)
4. Create Network graph (using the relationship among users to apply centrality measures )
5. Fuzzy clustering (assigning profiles to the users)

# Pictorial Representation

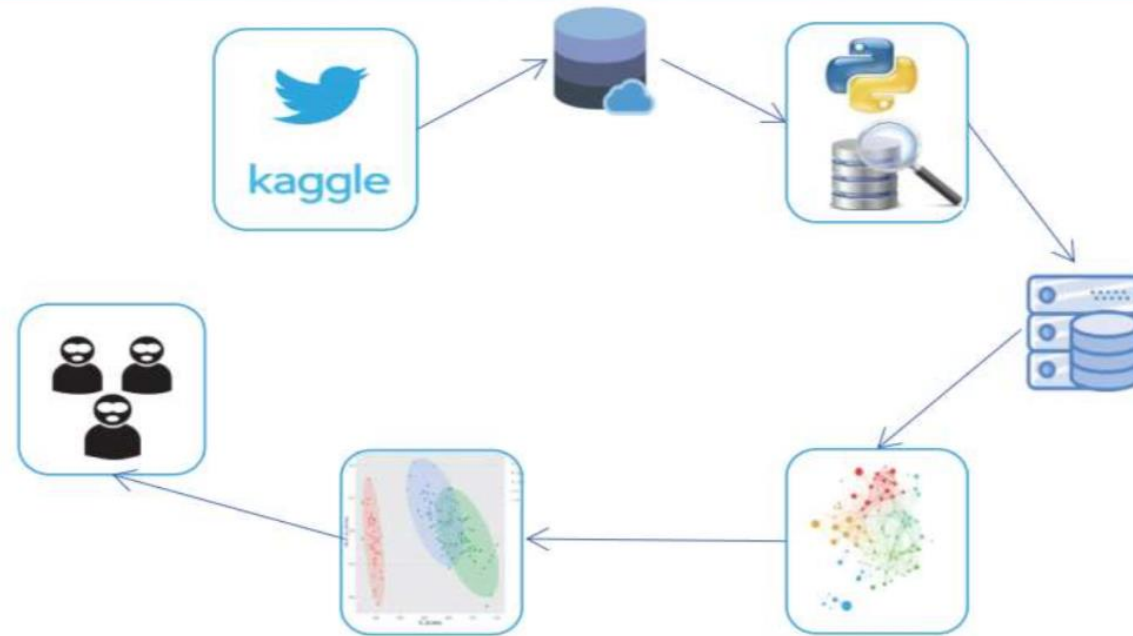
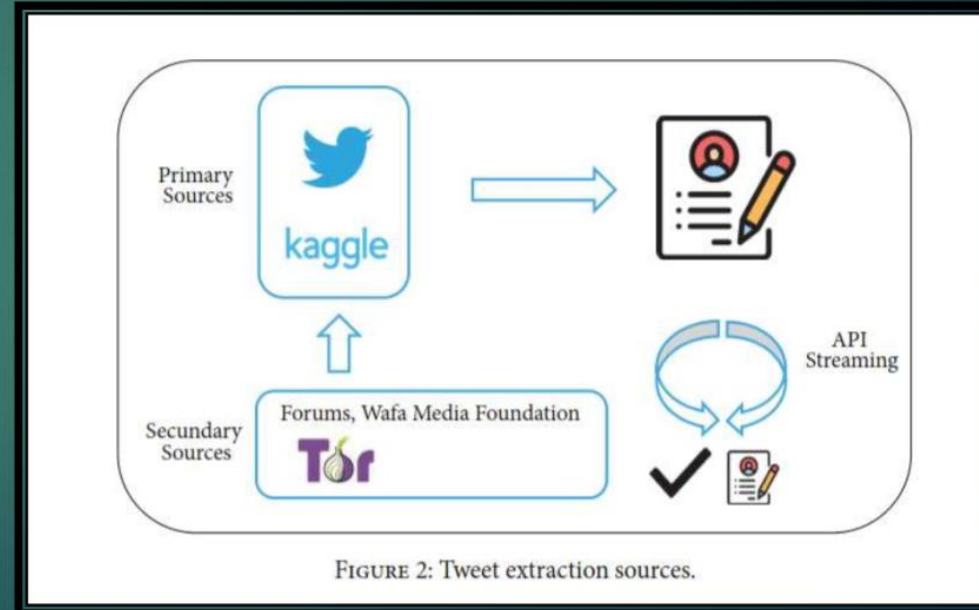


FIGURE 1: Stages of the procedure to select hidden Twitter profiles.

# Tweet extraction

## Different Sources

- ▶ **Primary source** (collect the base information of users actively spread terrorist contents)
- ▶ **Secondary sources** (get familiar with vocabulary used by user in social propaganda)





# Dataset After filtration of data

Name, Username, Description, Location, Number of followers (at the time the tweet was downloaded), Number of statuses by the user (when the tweet was downloaded), Date and timestamp of the tweet, the tweet itself

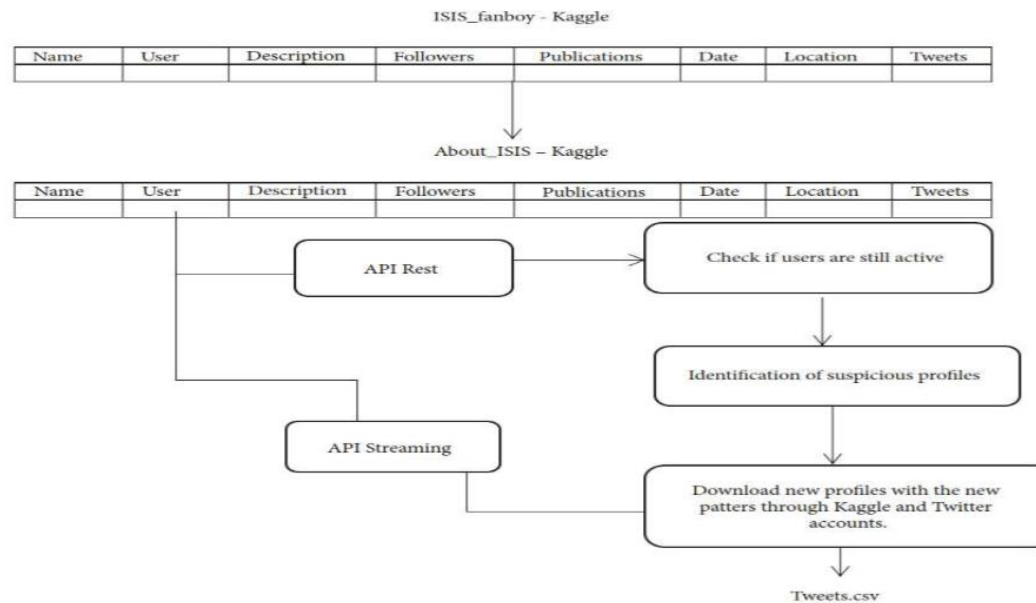


FIGURE 3: Tweet extraction procedure.



# Information Preprocessing

## Attributes

- ▶ Frequency (calculate interval of sending tweet)
- ▶ Sentiment (polarize the tweets in + and -)
- ▶ Regular expression (extract mentioned or retweet user)

# Sentiment analysis for input in fuzzy clustering

- ▶ Which clergy do pro-ISIS fanboys quote the most and which ones do they hate the most?
  - ▶ Search the tweet for names of prominent clergy and classify the tweet as positive, negative, or neutral accordingly the term used below

America, Christianity, Islam, Israel, Judaism, Mubarak, Palestine, Al-qaeda

- ▶ Classify each tweet as positive , negative and neutral
- ▶ This output feed to the fuzzy clustering for detecting dangerous suspicious node

# Social Network analysis

Adjacency Matrix of the graph of Active users

|            | @user<br>1 | @user<br>2 | @use<br>3 | @user<br>4 | @user<br>5 | @user<br>6 | @user<br>7 |
|------------|------------|------------|-----------|------------|------------|------------|------------|
| @user<br>1 | 0          | 5          | 1         | 86         | 8          | 43         | 4          |
| @user<br>2 | 5          | 0          | 46        | 8          | 9          | 45         | 83         |
| @user<br>3 | 99         | 75         | 0         | 11         | 10         | 45         | 42         |
| @user<br>4 | 64         | 9          | 64        | 0          | 8          | 6          | 8          |
| @user<br>5 | 9          | 89         | 46        | 4          | 0          | 75         | 7          |
| @user<br>6 | 11         | 3          | 4         | 5          | 1          | 0          | 12         |
| @user<br>7 | 36         | 7          | 6         | 75         | 3          | 56         | 0          |

Let  $G(v,e)$  graph create  
Adjacency matrix form from  
the active users

**Vertex** ( $v$ ) : number of Users

**Edges** ( $e$ ) : retweets by users /  
mention to a user /reply to a  
tweet

Weight( $w$ ) : no. of retweet or  
mention the name user inside  
a tweet

# Graphing the network centrality

- ▶ **Degree centrality measure** ( very connected users with relationship among others node)

$$C_{\text{deg}}(V) = \frac{\text{deg}(V)}{|N| - 1}$$

- ▶ **Closeness** : (how close to others node and They do not have to be very influential but are very active followers who broadcast information)

$$C(x) = \frac{(|N|)}{\sum_y d(y, x)}$$

- ▶ **Betweenness** : centrality(indicates the nodes included in the shortest path between most of the nodes. Low followers but connect many relevant nodes)

$$g(v) = \sum_{s \neq v \neq t} \frac{\sigma_{st}(v)}{\sigma_{st}}$$

Where  $\sigma_{st}$  is the total number of shortest paths from node s to node t.  
 $\sigma_{st}(v)$  is the number of those paths that pass through v.

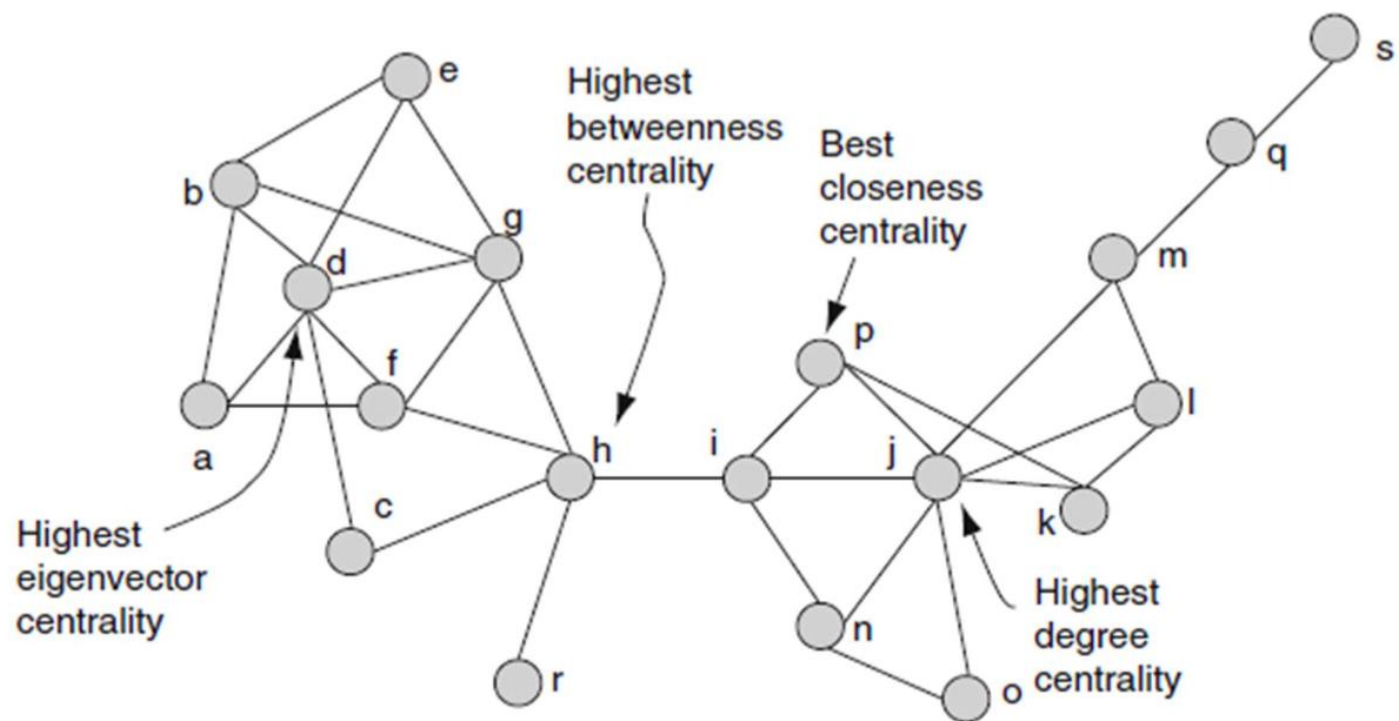
- ▶ **Eigen vector** : the number of links of a given user ,helps to computes not only your connection but the connection of whom you follow,

Let  $A=(a_{v,t})$  be the adjacency matrix and relative centrality  $v$  is given by,

$$x_v = \frac{1}{\lambda} \sum_{t \in M(v)} x_t = \frac{1}{\lambda} \sum_{t \in G} a_{v,t} x_t$$

- ▶ Where  $a_{v,t}=1$  if vertex  $v$  is linked to vertex  $t$  and  $a_{v,t}=0$  , otherwise
- ▶  $M(v)$  is a set of the neighbors of  $v$  ,and  $\lambda$  is a constant
- ▶ With small rearrangement this can be rewritten in vec.tor notation as the eigen vector equation

$$\mathbf{Ax} = \lambda \mathbf{x}$$





# Clustering use for classify influencers in profiles

## Fuzzy clustering:

used to classify information in groups created from similarities among individuals.

$$J(Z, U, V) = \sum_{i=1}^c \sum_{j=1}^N (\mu_{ij})^m \|z_j - v_i\|_A^2$$

c: no. of cluster

N: no. of data

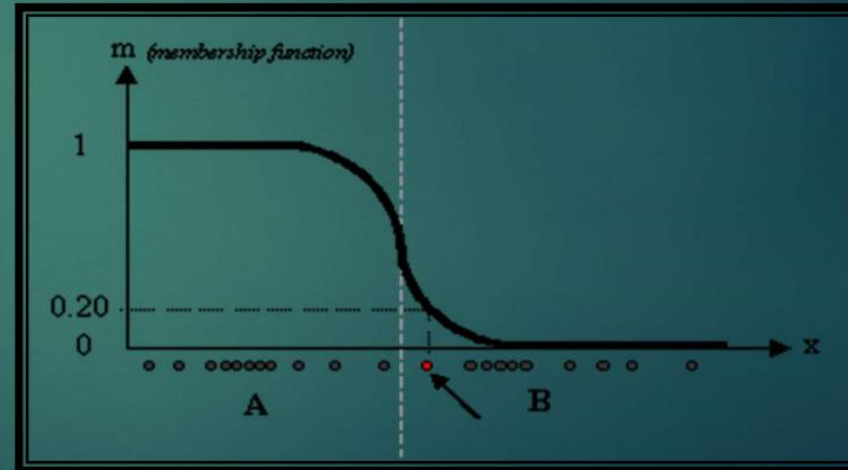
m : is any real number greater than 1

$U_{ij}$  : is the degree of membership of  $z_i$  in the cluster j

$z_i$  : is the i-th of dimensional measured data

$v_j$  : is the d-dimension center of the cluster

$\| * \|$  : is any norm expressing the similarity between any measure data and the center





## Continuing..

Where

Fuzzy partition matrix of  $Z$

$$\mathbf{U} = [\mu_{ik}] \in M_{fc}$$

Vector of cluster centres

$$\mathbf{V} = [\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_c], \quad \mathbf{v}_i \in \mathbb{R}^n$$

# Constraints and modification on this algorithm imposed

## Norm-Inducing matrix

which help to way with its determinant fixed corresponds to optimizing the cluster's shape while its volume remains constant

$$\mathbf{A}_i = [\rho_i \det(\mathbf{F}_i)]^{1/n} \mathbf{F}_i^{-1},$$

Where constraint  $|\mathbf{A}_i| = \rho_i, \rho_i > 0, \forall i.$

## Modification

1.adaptive distance measure for non spherical structure.

$$D_{ijA_i}^2 = (z_j - v_i)^T A_i (z_j - v_i)$$

2. Algorithms relaxing the condition on the probability of belonging to each segment indicating a level between each one of the groups

# Continuing..

Fuzzy covariance matrix  
of  $i$ th cluster

$$\mathbf{F}_i = \frac{\sum_{k=1}^N (\mu_{ik})^m (\mathbf{z}_k - \mathbf{v}_i)(\mathbf{z}_k - \mathbf{v}_i)^T}{\sum_{k=1}^N (\mu_{ik})^m}.$$

# Fuzzy parameters determine in algorithm

- ▶ Number of clusters (most important to divide the data )
- ▶ Fuzziness parameter ( $m$  effects significant to fuzziness in the segmentation )
- ▶ Termination criterion ( to stop the algorithm here 1000 iteration or reach error lower than 0.005 )
- ▶ Distance matrix (the calculation of distance implies establishing the scalar product matrix)

# Gustafson-Kessel Algorithm

The objective function of GK algorithm defined by

$$J(\mathbf{Z}; \mathbf{U}, \mathbf{V}, \{\mathbf{A}_i\}) = \sum_{i=1}^c \sum_{k=1}^N (\mu_{ik})^m D_{ik\mathbf{A}_i}^2$$

# Input data matrix

Each observation consists of  $n$  measured variables, grouped into an  $n$ -dimensional column vector  $z_k = [z_{1k}, \dots, z_{nk}]^T$ ,  $z_k \in \mathbb{R}^n$ . A set of  $N$  observations is denoted by  $Z = \{z_k \mid k = 1, 2, \dots, N\}$  and is represented as an  $n \times N$  matrix.

$$Z = \begin{bmatrix} z_{11} & z_{12} & \cdots & z_{1N} \\ z_{21} & z_{22} & \cdots & z_{2N} \\ \vdots & \vdots & \vdots & \vdots \\ z_{n1} & z_{n2} & \cdots & z_{nN} \end{bmatrix}$$

No. of observation

no. of variables

$n \times N$

| user                         | Us<br>er<br>1 | us<br>er<br>2 | us<br>er<br>3 | ..... | Us<br>er<br>n |
|------------------------------|---------------|---------------|---------------|-------|---------------|
| feature                      |               |               |               |       |               |
| frequency                    |               |               |               |       |               |
| Sentiment<br>result          |               |               |               |       |               |
| Eigen<br>vector              |               |               |               |       |               |
| @ mention<br>Name of<br>user |               |               |               |       |               |



# Gustafson – Kessel algorithm

Given the data set  $\mathbf{Z}$ , choose the number of clusters  $1 < c < N$ , the weighting exponent  $m > 1$  and the termination tolerance  $\epsilon > 0$  and the cluster volumes  $\rho_i$ . Initialize the partition matrix randomly, such that  $\mathbf{U}^{(0)} \in M_{fc}$ .

**Repeat for**  $l = 1, 2, \dots$

**Step 1:** Compute cluster prototypes (means):

$$\mathbf{v}_i^{(l)} = \frac{\sum_{k=1}^N \left( \mu_{ik}^{(l-1)} \right)^m \mathbf{z}_k}{\sum_{k=1}^N \left( \mu_{ik}^{(l-1)} \right)^m}, \quad 1 \leq i \leq c.$$

**Step 2:** Compute the cluster covariance matrices:

$$\mathbf{F}_i = \frac{\sum_{k=1}^N \left( \mu_{ik}^{(l-1)} \right)^m (\mathbf{z}_k - \mathbf{v}_i^{(l)}) (\mathbf{z}_k - \mathbf{v}_i^{(l)})^T}{\sum_{k=1}^N \left( \mu_{ik}^{(l-1)} \right)^m}, \quad 1 \leq i \leq c.$$

**Step 3:** Compute the distances:

$$D_{ik\mathbf{A}_i}^2 = (\mathbf{z}_k - \mathbf{v}_i^{(l)})^T \left[ \rho_i \det(\mathbf{F}_i)^{1/n} \mathbf{F}_i^{-1} \right] (\mathbf{z}_k - \mathbf{v}_i^{(l)}), \\ 1 \leq i \leq c, \quad 1 \leq k \leq N.$$

**Step 4:** Update the partition matrix:

for  $1 \leq k \leq N$   
if  $D_{ik\mathbf{A}_i} > 0$  for all  $i = 1, 2, \dots, c$

$$\mu_{ik}^{(l)} = \frac{1}{\sum_{j=1}^c (D_{ik\mathbf{A}_i} / D_{jk\mathbf{A}_i})^{2/(m-1)}},$$

**until**  $\|\mathbf{U}^{(l)} - \mathbf{U}^{(l-1)}\| < \epsilon$ .

# Fuzzy partition matrix

$$\mathbf{U} = [\mu_{ik}] \in M_{fc}$$

Initial condition

| <b>Data<br/>Cluster</b> | <b>z1</b> | <b>z2</b> | <b>.....</b> |  |  |  | <b>zk</b> |
|-------------------------|-----------|-----------|--------------|--|--|--|-----------|
| c1                      | 1         | 0         |              |  |  |  | 0         |
| c2                      | 0         | 1         |              |  |  |  | 1         |

Where,  
i represent cluster  
number  
k represent the data

c\* N

## 4.Result

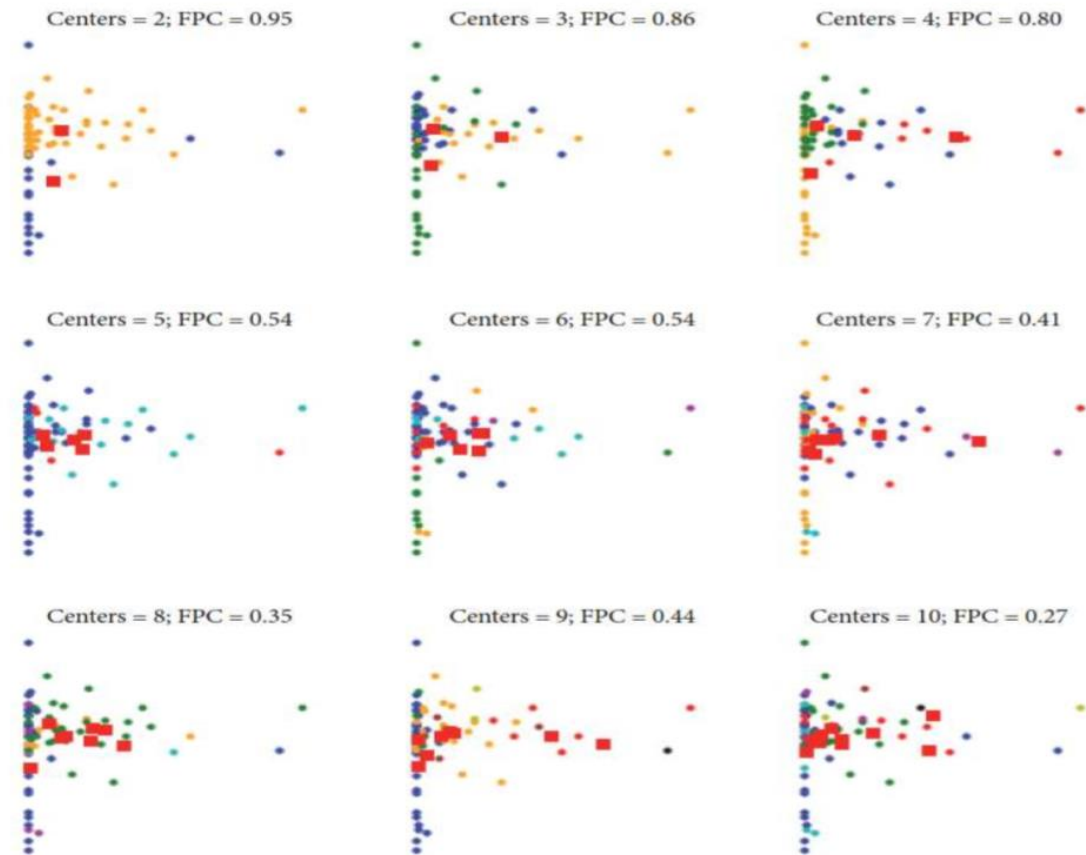


FIGURE 6: Results of the pro-Isis users with different levels of segmentation.

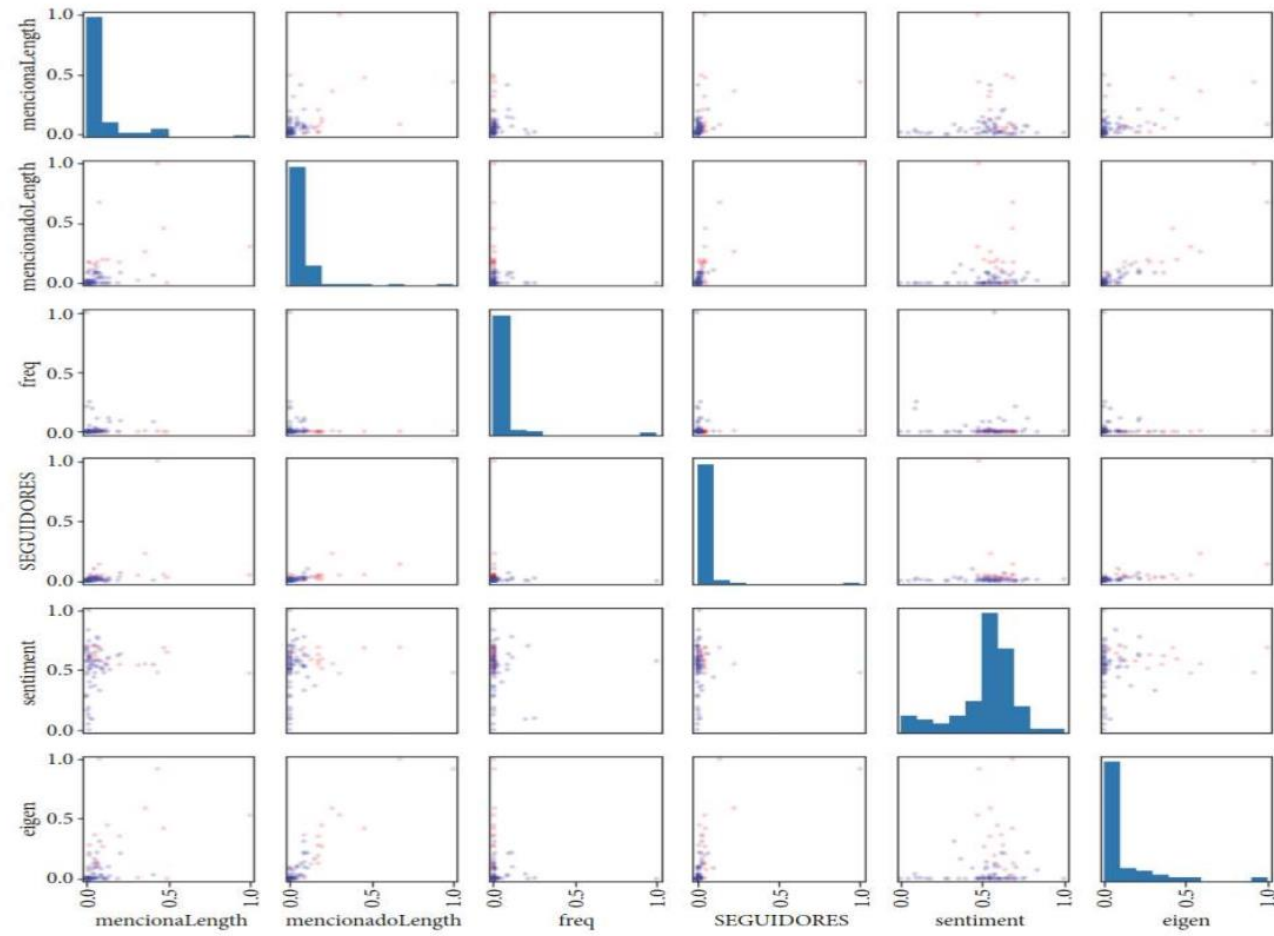


FIGURE 7: Distribution of the variables used in the fuzzy clustering.



## 5. Data availability

- ▶ Kaggle repository under the name How ISIS Uses Twitter

<https://www.kaggle.com/fifthtribe/how-isis-uses-twitter>

# Some Application

- ▶ Human resources departments can apply the power of profile identification to automatically classify candidates
- ▶ security teams can detect undesirable clients in the financial or insurance sectors.
- ▶ To identify the fake news about any company to damage economical damages.
- ▶ Political manipulation of significant number of voter by the influencers.





Thank You